

Encapsulated Ionic Liquids with MOF-Driven CO₂ Channels: Overcoming Kinetic Limits for Rapid Carbon Capture

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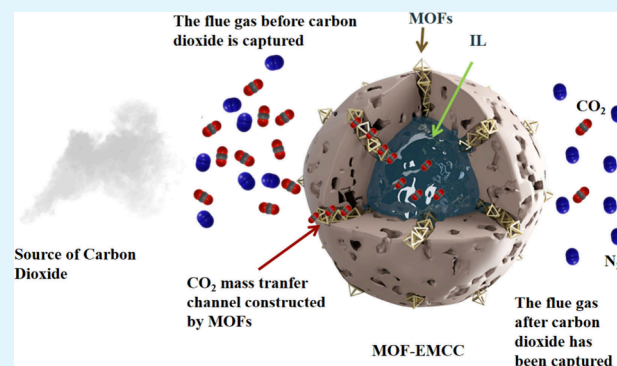
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ABSTRACT: The urgent challenge of anthropogenic climate change has intensified demand for innovative carbon capture solutions. However, conventional CO₂ capture technologies such as adsorption, absorption, and membrane separation are limited by high energy costs and scalability limitations, which restrict their viability as future-oriented solutions. This study introduces a novel microcapsule capture system (MECS) that synergizes liquid-phase selectivity and solid-phase mass transfer efficiency, overcoming kinetic bottlenecks from the low specific surface area and high viscosity of ionic liquids. The ionic liquid [Bmim][Ac] was encapsulated within a sodium alginate gel matrix. This strategy significantly enhances the interfacial contact area with CO₂, which accelerates capture kinetics. The spatial confinement within microcapsules mitigates viscosity-induced kinetic bottlenecks that improves adsorption capacity. Further, adding a zirconium-based metal–organic framework (UiO-66-NH₂) into the shell creates engineered CO₂ transport channels and reduces diffusion resistance and increases the capture rate ~5 times. Under simulated flue gas, UiO-66-NH₂-embedded microcapsules achieved a CO₂ adsorption capacity of 1.62 mmol/g and surpassed that of pure [Bmim][Ac] (0.99 mmol/g). They demonstrated CO₂/N₂ selectivity of 28 and exceptional mechanical resilience, withstanding loads up to 20000 times their mass. Thus, under typical industrial conditions, the microcapsules show superior stability, capture capacity, and efficiency, which makes them well-suited for large-scale carbon capture.

KEYWORDS: Carbon Capture, Microcapsule, Extrusion Vibration, Ionic Liquid, MOFs



INTRODUCTION

Carbon capture, utilization, and storage (CCUS) plays a pivotal role in accelerating progress toward the strategic goal of “carbon neutrality” to mitigate climate change impacts. Within the CCUS framework, CO₂ capture represents the most substantial cost, accounting for over 70% of the total process expenses, and remains the primary bottleneck limiting the broad implementation of CCUS technologies.^{1–4} CO₂ capture technologies typically include absorption, adsorption, and membrane separation. Monoethanolamine (MEA) is a prevalent absorber utilized for CO₂ separation following combustion processes in thermal power plants.^{5–8} Other notable absorbers, such as organic amine absorbers, amine mixtures, and ionic liquids, have also been widely studied.^{9–12} Nonetheless, these materials are generally characterized by high operational costs, significant energy requirements, and limited capture rates. Additionally, adsorption is a widely recognized method for CO₂ capture, employing materials like porous carbon, zeolite, silica, and metal–organic frameworks, which are often discussed in the literature.^{13–16} Despite its advantages in capture efficiency and rate, high sensitivity to water adsorption, and limited lifespan. The full potential of adsorption technology has yet to be realized.^{17–20} Further-

more, membrane separation offers substantial benefits in terms of energy efficiency, operational simplicity, and environmental sustainability. However, the inherent trade-off between permeability and selectivity in traditional polymer membranes undermines their industrial viability.^{21,22} Therefore, the development of innovative, cost-effective, and efficient CO₂ capture technologies remains crucial for enabling the low-cost, large-scale deployment of CCUS technologies.

The development of new carbon capture technologies seeks to address the myriad challenges presented by traditional methods. Among these, microcapsule carbon capture systems (MECS) emerge as a promising near-term solution, effectively combining the extensive surface area of solid adsorbents with the selectivity of liquid adsorbents. This integration facilitates rapid and controlled absorption and release of CO₂ throughout the carbon cycle.^{23–27} Initially proposed by Vericella et al. in

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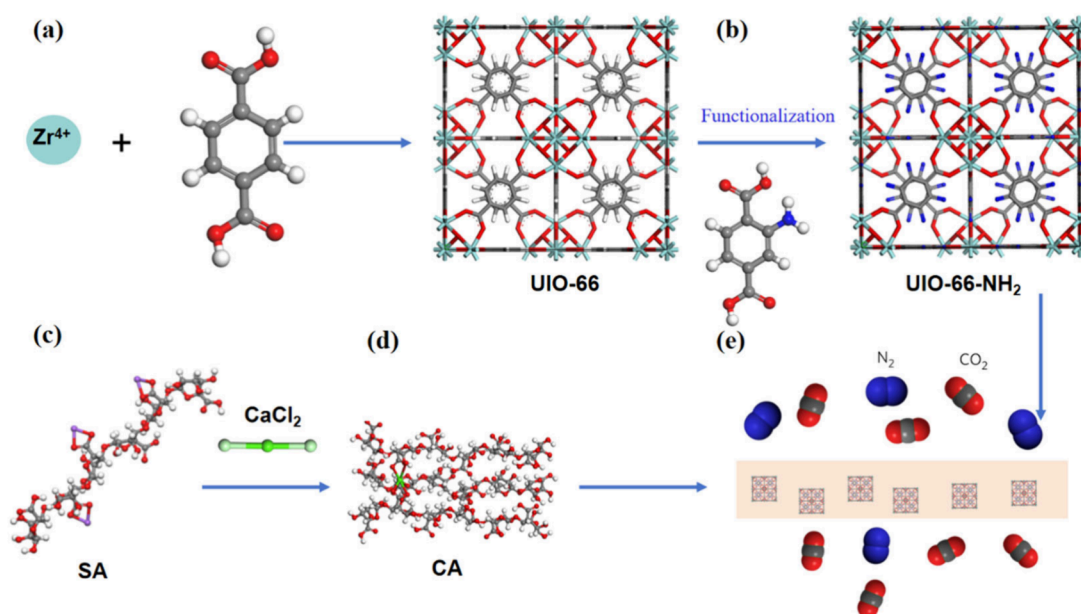


Figure 1. Schematic diagram of microcapsule solidified shell introducing MOFs. (a, b) UiO-66-NH₂ synthesized from zirconium ion and 2-aminoterephthalic acid. (c) Molecular structure of SA. (d) SA sol cured in a water bath in CaCl₂ solution to produce CA gel. (e) MOF-EMCC shell prepared by CA gel doped with MOFs for carbon capture. Carbon atoms are depicted in gray; oxygen and hydrogen atoms are shown in red and white, respectively; nitrogen, -NH₂ and bromine are represented in blue.

2015, microcapsule carbon capture technology utilizes microfluidic assembly techniques to fabricate polymer microcapsules, comprising liquid carbonate absorbers (e.g., Na₂CO₃ solution) within rigid silicon shells, demonstrating enhanced carbon capture capabilities.²⁸ While pioneering, this approach presented inherent limitations for practical deployment. The impermeable silicon shell introduced significant CO₂ mass transfer resistance, hindering kinetics despite the encapsulated liquid's capacity. Furthermore, the rigid shell material posed challenges for scalable production and resulted in microcapsules with limited mechanical resilience, susceptible to fracture under operational stresses. Despite significant interest from researchers, these critical hurdles, notably the inherent CO₂ mass transfer resistance within the microcapsule shell imposed by dense polymer/silica materials, and challenges in large-scale production alongside insufficient mechanical properties, have restricted the potential benefits of this technology.

In this study, a novel MOF-doped carbon capture microcapsule was successfully developed using emulsion volumetric templating (EVT). These shell–core structured microcapsules synergize the advantages of liquid solvents, with their high capacity, selectivity, and water resistance, and solid adsorbents, which are known for their high surface area and low volatility. The microcapsule design substantially enhances the gas–liquid contact area and addresses major challenges associated with high-viscosity ionic liquids and other volatile, toxic liquid absorbers that conventional carbon capture systems face in large-scale power plant applications.

More critically, the incorporation of a UiO-66-NH₂ into the solidified shell significantly improves both the carbon capture capacity and efficiency of microcapsules (Figure 1). The selection of 2-aminoterephthalic acid as the organic ligand for UiO-66-NH₂ synthesis is pivotal: the introduced -NH₂ group acts as a Lewis basic site and enables strong dipole–quadrupole interactions and hydrogen bonding with CO₂,

thus enhancing adsorption capacity and selectivity.^{16,28} Additionally, the -NH₂ group strengthens the hydrolytic stability of the framework by reinforcing Zr–O bonds, which ensures durability in humid flue gas environments. The UiO-66-NH₂ material itself exhibits exceptional carbon capture capabilities, which in turn enhances the overall performance of the microcapsules. Furthermore, this material not only substantially strengthens the microcapsules' mechanical properties but also enriches the pore structure, expands the CO₂ mass transfer channels, and decreases mass transfer resistance.

Specifically, we developed carbon capture microcapsules featuring a novel and highly permeable shell. The EVT method facilitates rapid and large-scale production of microcapsules, significantly reducing manufacturing costs, which is crucial for the scaling and commercialization of carbon capture technologies. The MOFs-doped shell–core motif microcapsules overcome the principal challenges of traditional carbon capture technologies and offer a more cost-effective process compared to MEA. Unlike membrane separation, where permeability and selectivity often trade off, these microcapsules allow for independent optimization of adsorbent selectivity and permeability.

Additionally, compared to porous adsorbent materials, this new system addresses issues related to CO₂ selectivity, resistance to water vapor, and challenges associated with handling fine powders. The main drawbacks of [Bmim][Ac], such as the low specific surface area of high-viscosity liquids and slow CO₂ absorption/desorption kinetics, are mitigated through encapsulation in a MOF-alginate system. Although other types of capture media can also encapsulate carbon capture solvents like MEA and carbonates through MOFs-doped shell–core architectures, our initial demonstration focused on high-viscosity ionic liquids.

METHODS

Synthesis of Zr-MOFs (UiO-66-NH₂). UiO-66-NH₂ was synthesized using a conventional method, with the molar ratios of ZrCl₄, 2-aminoterephthalic acid, and *N,N*-dimethylformamide (DMF) set at 2.3 mmol, 2.3 mmol, and 30 mL, respectively. Initially, ZrCl₄ and 2-aminoterephthalic acid were dissolved in DMF and stirred for 1 h to prepare the precursor solution. The selection of 2-aminoterephthalic acid as the organic ligand for UiO-66-NH₂ synthesis was driven by its -NH₂ functional group, which enhances CO₂ adsorption via Lewis acid–base interactions and improves framework stability in humid environments, as validated in prior studies.^{16,28}

The mixture was subsequently transferred into a Teflon-lined stainless steel autoclave and subjected to hydrothermal crystallization at 120 °C for 24 h. For comparison, nonfunctionalized UiO-66 was synthesized following the standard procedure reported by Cavka et al.,¹⁶ using terephthalic acid as the organic ligand instead of 2-aminoterephthalic acid. All other steps (e.g., crystallization, washing, drying) were identical to those described for UiO-66-NH₂. The resulting UiO-66-NH₂ crystals were isolated by centrifugation and washed three times alternately with methanol and deionized water to remove excess unreacted ligands and solvents. The samples were then dried at 180 °C for 12 h and stored in a drying cabinet. In the final step, the powder sample was ultrasonically washed with deionized water for 3 days, with the deionized water being replaced daily, which ensured that the sample was dispersed in a colloidal form in the deionized water, suitable for microcapsule preparation.

Preparation of Zr-MOFs-Based Sodium Alginate (SA) Composites. To prepare Zr-MOFs-based SA composites, 2 wt % sodium alginate (SA) was first dissolved in deionized water and stirred for 1 h to obtain a sodium alginate hydrosol. To prevent particle agglomeration, the previously prepared MOF colloidal solution was subjected to ultrasonic treatment. Subsequently, the MOF colloidal solution and SA hydrosol were thoroughly mixed. After ultrasonic degassing to remove air bubbles, the Zr-MOFs-based SA composites were obtained by incorporating MOFs at varying concentrations of 0, 30, 40, 50, 60, and 70 wt % relative to pure SA base (note: MOF concentration refers to the weight ratio of MOFs to pure SA, not to the total mass of the SA sol).

Preparation of Microcapsules Using High-Frequency Extrusion Vibration. Microcapsules were prepared using a high-frequency extrusion vibration device, which incorporated a pressure bottle, an injection pump, a high-frequency vibration control unit, a concentric nozzle, and a curing unit. As shown in Supporting Information Figure S2, the prepared Zr-MOFs-based SA, loaded with MOFs at concentrations of 0, 30, 40, 50, 60, and 70 wt %, was introduced from the pressure bottle into the external annular channel of the concentric nozzle under controlled pressure. Simultaneously, the ionic liquid was delivered into the central channel of the nozzle via the injection pump to generate a laminar jet and combine with the shell fluid at the microcapsule nozzle. This laminar flow was sheared into shell–core microcapsule droplets under high-frequency vibration. The droplets were then solidified upon entering a 2% CaCl₂ solution water bath, forming a solid shell. After curing, the microcapsules were freeze-dried at low temperatures. During this process, the internal water sublimated into gas that preserved their structural integrity, minimized shrinkage, and enhanced the pore structure through controlled ice crystal formation. Furthermore, prior to freeze-drying, the microcapsules underwent a series of pretreatment steps, which included cleaning and screening to remove impurities and ensure the stability of the microcapsules. Microcapsule samples with 0 wt % MOF loading were designated SA-MCC, while those loaded with 30 wt % to 70 wt % MOFs were referred to as MOF-EMCC-*y*, where *y* denotes the MOF loading percentage. Unless otherwise specified, MOF-EMCC refers to MOF-EMCC-60 throughout the following text.

Characterization. A variety of characterization techniques were employed in this research to elucidate the physical and chemical properties of MOF-enhanced carbon capture microcapsules. Initially, CO₂ adsorption capacity was measured using a physical adsorption analyzer (ASAP 2020, Micromeritics Instrument, USA). Samples were

activated under vacuum at 180 °C for 12 h, and helium was used at -196 °C to measure free space, ensuring test accuracy and repeatability. CO₂ adsorption isotherms were obtained over a pressure range from 10⁻⁶ to 1 bar. Additionally, to assess carbon capture performance under realistic conditions, microcapsules were tested using a multicomponent breakthrough adsorption analyzer (BSD-MAB, Beishide Instrument, China) simulating coal flue gas environments. Further, microcapsules, identified as MOF-EMCC-*y* or SA-MCC, underwent thermogravimetric analysis (TGA) using a thermogravimetric analyzer (Q500, TA Instruments, USA). This analysis involved gradual heating to study their thermal stability and decomposition characteristics under typical coal flue gas conditions, which ensured stable and reliable operation in these environments.

The mechanical stability and fragmentation resistance were evaluated using a particle strength meter (SKZ-500, Xiangyi Instrument, China), which provided insights into the mechanical properties of microcapsules in moving bed or vulcanized bed configurations during flue gas carbon capture. Furthermore, Surface morphology and microstructure were observed using a scanning electron microscope (SEM) equipped with energy-dispersive X-ray (EDX) detection (Evo LS 15, Zeiss, Germany; XFlash 5010, Bruker, Germany). SEM analysis was performed under vacuum with a 3 kV acceleration voltage and 10000× magnification. The qualitative and quantitative analysis of elemental composition was performed through EDX examination.

Finally, X-ray diffraction (XRD) analysis was performed using a Bruker D2 Phaser with Cu Kα₁ radiation (wavelength of 1.54060 Å). The Scherrer equation and Bragg's law were applied to investigate the crystal structure and grain size of the samples. These comprehensive characterization methods provided extensive data on the microcapsule samples, chemical composition, physical and chemical properties, thus offering a vital reference for further research and application.

Computational Research. To achieve a comprehensive understanding of the CO₂ mass transfer process in MOFs-enhanced microcapsules, a series of simulations were conducted using density functional theory (DFT). These calculations were performed using the Vienna Ab-initio Simulation Package (VASP). This approach enabled the detailed examination of molecular interactions and transport mechanisms at the atomic level. This provided valuable insights into the efficiency and dynamics of CO₂ capture within the microcapsules. The Perdew–Burke–Ernzerhof (PBE) functional within generalized gradient approximation (GGA) framework was used to describe the exchange–correlation effects, while the projector-augmented wave (PAW) pseudopotential was applied with a kinetic energy cutoff of 500 eV, which was utilized to describe the expansion of the electronic eigenfunctions. A vacuum thickness of 25 Å was set to minimize interlayer interactions. Brillouin-zone integration was sampled using a Γ -centered $5 \times 5 \times 1$ Monkhorst–Pack *k*-point. All atomic positions were fully relaxed until the energy and force converged below 1×10^{-5} eV and 0.03 eV/Å, respectively. To account for long-range interactions, the dispersion-corrected DFT-D method was employed. Finally, the climbing image nudged elastic band (CI-NEB) method was used to compute the minimum energy pathway of the cyclization reaction along with its activation energy barrier.

The adsorption energy (E_{ads}) of a complex formed between two molecules, A and B, can be calculated using the following equation:

$$E_{\text{ads}} = E_{\text{complex}} - (E_{\text{A}} + E_{\text{B}})$$

where E_{complex} is the total energy of the molecular complex of A and B. E_{A} and E_{B} are the total energies of isolated molecules A and B, respectively.

RESULTS

Design and Fabrication of Microcapsules. The microcapsules containing ionic liquids are prepared using the EVT method, as illustrated in Figure S1. The process involved injecting the shell solution into the microcapsule nozzle via a pressure bottle. Concurrently, the internal fluid was propelled

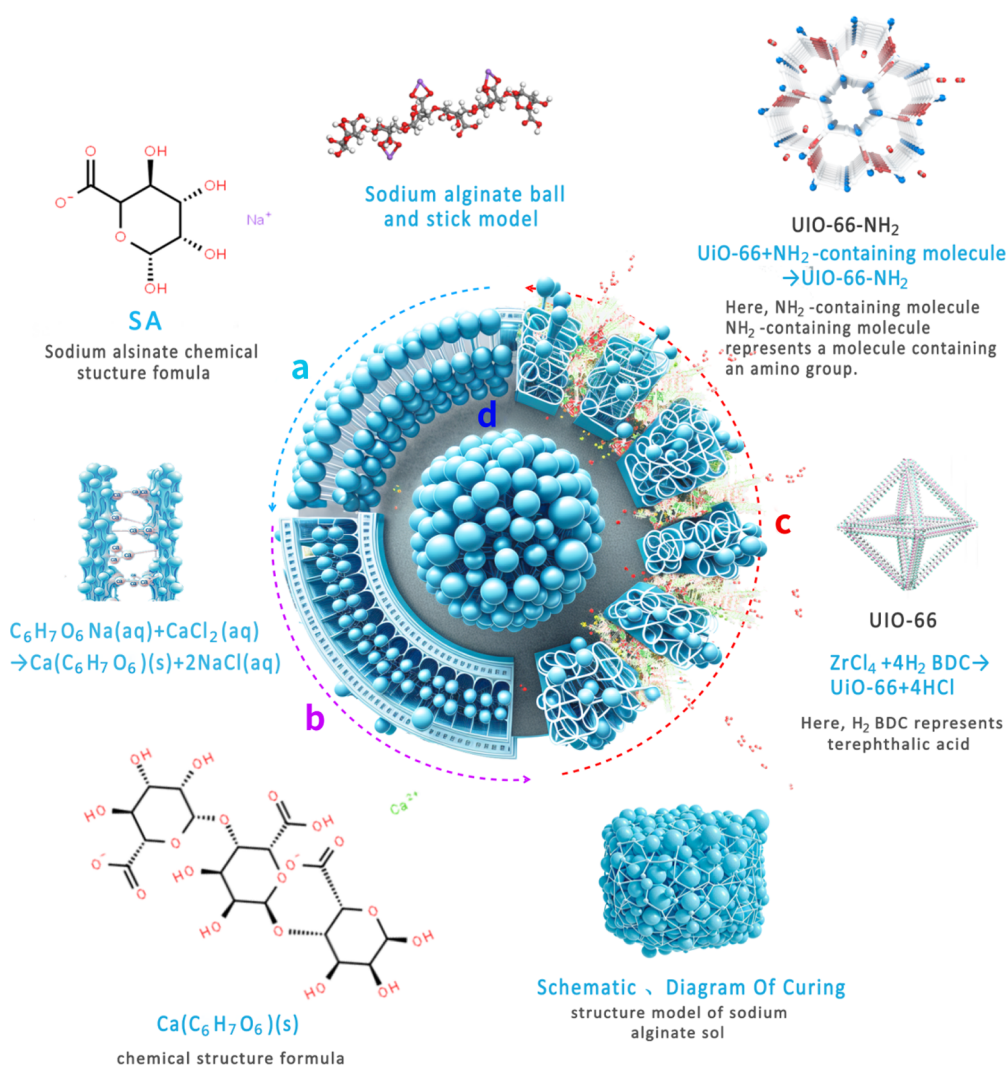


Figure 2. Microcapsules for carbon capture. (a) SA-MCC microcapsules droplet with shell fluid as sodium alginate sol. (b) Sol–gel curing reaction process of SA-MCC microcapsules droplet. (c) As comparison with SA-MCC, MOF-EMCC- γ introduction of MOF into sodium alginate sol to construct the transport path of CO₂. (d) Schematic diagram of ionic liquid (internal fluid).

into the nozzle by an injection pump, which resulted in the formation of laminar flows of the internal and shell fluids at the nozzle. A high-frequency vibration control unit then sheared this laminar jet fluid into shell–core microcapsule droplets. The vibration frequency controlled the droplet formation rate; for instance, a frequency of 700 Hz generated approximately 700 droplets per second, reaching a maximum tested rate of ~3000 droplets per second with a single nozzle under optimized conditions. This translates to a conservative production capacity of >250000 microcapsules per hour per nozzle (based on 700 Hz). Critically, the EVT process is inherently scalable through nozzle parallelization (e.g., multi-nozzle arrays), straightforward process intensification (e.g., wider nozzles for larger capsules), and continuous operation. This potential for high throughput (>10⁶–10⁷ capsules/h with moderate parallelization) contrasts sharply with the typically low production rates (often <<1000 capsules/h) associated with conventional microfluidic techniques used in earlier encapsulation studies,²⁸ positioning EVT as a viable route for industrial-scale manufacturing. These droplets entered a water bath curing unit to solidify into a stable shell.

The carbon capture microcapsules were constituted of two primary components: (1) the internal fluid, consisting of an ionic liquid, and (2) the shell fluid, comprising a UIO-66-NH₂ and sodium alginate composite sol. The preparation processes for SA-MCC and MOF-EMCC- γ and their underlying principles are illustrated in Figure 2.

Figure 2a demonstrates the formation of SA-MCC microcapsule droplets, where sodium alginate serves as the shell fluid. Sodium alginate molecules, rich in carboxyl and hydroxyl groups, formed hydrogen bonds with water molecules. This interaction facilitated the random structuring between molecules and maintained the sodium alginate in a sol state and created a robust interfacial force on the ionic liquid surface. Consequently, the sodium alginate sol uniformly envelops the ionic liquid, culminating in a well-defined microcapsule droplet structure, as shown in Figure S3a.

Figure 2b illustrates the curing reaction when SA-MCC droplets entered a calcium chloride solution. Here, a rapid ion exchange reaction occurs, with calcium ions from the solution replacing sodium ions in the sodium alginate, thus forming cross-linking bonds between the calcium ions and alginate molecules. These bonds knit the alginate molecules into a

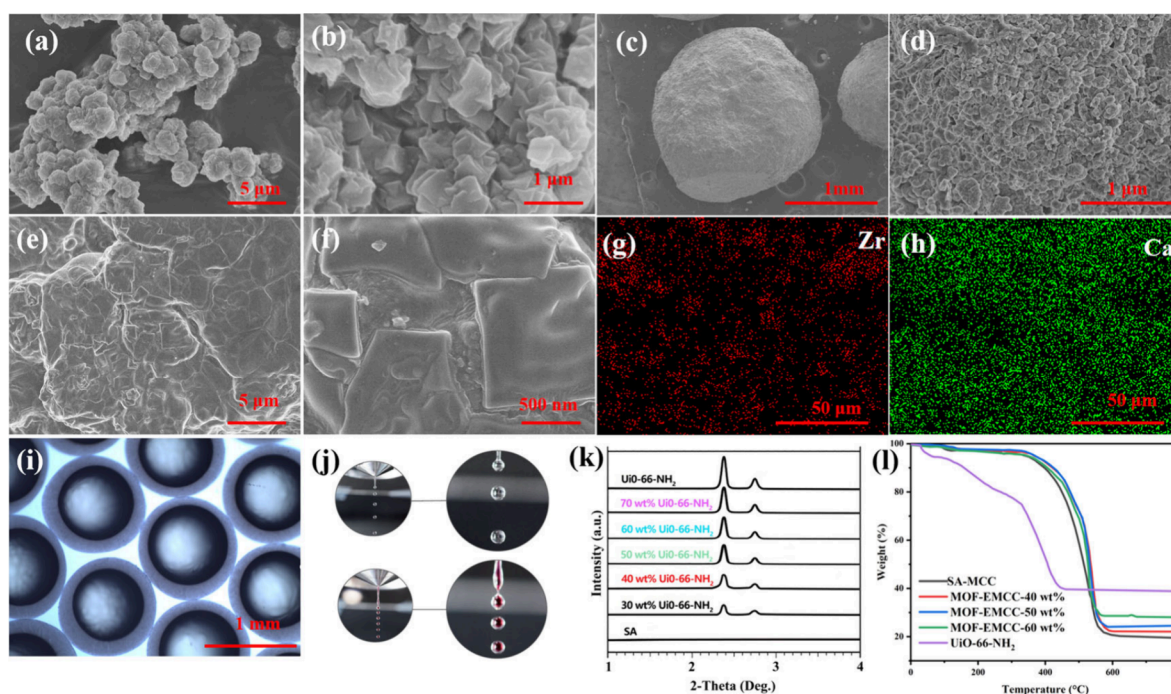


Figure 3. MOF and MOF-EMCC characterization. (a, b) SEM images of dried aggregates of micro-sized UiO-66-NH₂. (c, d) SEM morphology of SA-MCC. (e, f) Magnified features of the surface of MOF-MCC. (g, h) X-ray energy spectrum element mapping of Zr and Ca elements, respectively, for MOF-EMCC. (i) Optical microscope image of SA-MCC. (j) Jet fluid of the concentric nozzle forming a laminar flow of shell fluid and inner fluid, which is uniformly sheared into microcapsule droplets under the action of high-frequency vibration. (k) XRD diagram of MOF-EMCC. (l) TGA curve of carbon capture microcapsule.

three-dimensional network, and transform the fluid into a gel state, as depicted in Figure S3b, which illustrates the fully cross-linked microcapsule structure.

Furthermore, Figure 2c, relative to SA-MCC, demonstrates how MOF-EMCC-y incorporates MOFs into the sodium alginate sol to enhance CO₂ transport pathways. Due to the crystalline and porous structure of MOFs, CO₂ molecules diffuse more efficiently than in traditional polymer matrices. The embedded MOF particles create dedicated CO₂ diffusion channels, complementing the microporous network of the surrounding polymer, as shown in Figure S3c, which depicts the fully integrated MOF-SA-MCC structure. The size of the prepared microcapsules ranged from 1 mm to 20 mm, with the droplet sizes approximately double the diameter of the selected nozzle. During the microcapsule fabrication process, the inner and shell fluid flow rates were independently controlled. By adjusting these flow velocities, the wall thickness of the shell material and the overall size of the microcapsules were precisely regulated. Importantly, the use of concentric nozzles with adjustable diameters enabled flexible tuning of microcapsule dimensions across a wide range. This method facilitated the production of monodisperse microcapsules with an encapsulation efficiency approaching 99%, which were well-suited for a variety of fluid viscosities, flow rates, and vibration frequencies.

Characterization and Mechanical Stability. We developed an efficient and straightforward synthesis method for MOF-EMCC microcapsules. As illustrated in Figure 3j, the shell and inner fluids converged at the concentric nozzle to form a laminar jet, which was then sheared into uniform microcapsule droplets by high-frequency vibration. Upon introduction into the calcium chloride solution in the curing unit, the shell solidified upon entry into the CaCl₂ solution.

Figure 3i presents optical microscope images of the fabricated MECS, which demonstrate highly uniform wall thicknesses (2 ± 0.5 mm) and diameters (10 ± 2 mm) with consistent centrality. The dimensions of the microcapsules were finely tuned by adjusting the flow rates of the inner and shell fluids. Specifically, shell thickness was inversely proportional to the inner fluid flow rate and directly proportional to the shell fluid flow rate.

Significantly, these geometries facilitated an increase in the adsorbent surface area by 1–2 orders of magnitude per volume of the reactor compared to the liquid in a typical support filler, as indicated in Figure S5.²⁹ The microcapsules, after being cured in a water bath, were subjected to freeze-drying to sublimate the internal water content into a gas, thus preserving their structural integrity against substantial shrinkage. Postdrying, the microcapsules displayed minor deformation in SEM images and maintained a regular spherical shape, as illustrated in Figure 3c. In contrast, Figure 3d shows the SEM image of SA-MCC, which exhibits a dense stacked structure. This stacking may hinder CO₂ mass transfer and result in compromised mechanical performance and its effectiveness in carbon capture applications.

UiO-66-NH₂ is a crystalline metal–organic framework consisting of organic ligands and metal ions, characterized by a highly ordered porous structure and excellent potential in gas adsorption. The particles of UiO-66-NH₂ manifest as agglomerated cubic crystals ranging from 500 to 1000 nm, as demonstrated in Figure 3a, with the MOF exhibiting a distinct polyhedral shape evident in Figure 3b. Besides, images of loaded MOFs microcapsules (MOF-EMCC) show a uniform and smooth surface structure, with a discernible rich MOFs structure at both lower and higher magnifications, as shown in Figure 3e,f, respectively. The incorporation of MOFs

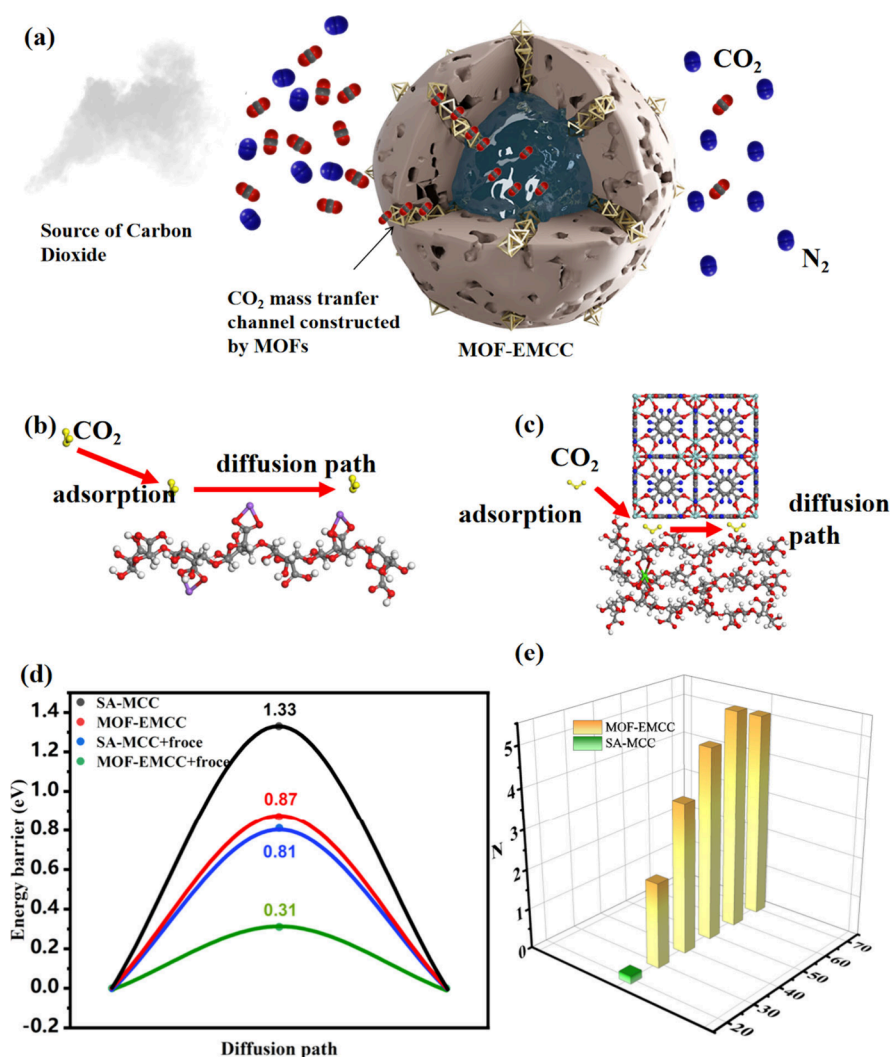


Figure 4. (a) Schematic of CO₂ diffusion pathways in MOF-EMCC, highlighting MOF channels (golden) and sodium alginate matrix (gray). (b, c) DFT-simulated CO₂ diffusion trajectories in MOF-EMCC. (d) Diffusion energy barriers for CO₂ in SA-MCC (black and blue) and MOF-EMCC (red and green), calculated via CI-NEB method. (e) Mechanical properties of SA-MCC and MOF-EMCC.

significantly enhances the mechanical properties of the microcapsules due to strong interface interactions and the effects of nanofillers, as verified by subsequent mechanical properties tests. Further, EDX spectroscopy was employed to analyze the loaded MOFs microcapsules, which confirmed the uniform distribution of MOFs material, as evidenced by the characteristic elements of Zr-MOFs in Figure 3g,h.

Microcapsules were prepared with UiO-66-NH₂ loadings ranging from 30 wt % to 70 wt %, yielding capsules with diameters between 1 mm and 10 mm. Powder XRD analysis of the microcapsule shells confirmed MOF nanocrystals' structural integrity at all loading levels, as shown in Figure 3k. Due to its inherently amorphous nature, pure sodium alginate gel did not exhibit the typical diffraction peaks. However, all microcapsule shells containing UiO-66-NH₂ displayed at least two significant diffraction peaks at 2θ values of 7.247° and 8.411°, verifying that UiO-66-NH₂ retained its crystallinity throughout the microcapsule fabrication process.^{30,31}

The thermogravimetric analysis (TGA) results for SA-MCC and MOF-EMCC are presented in Figure 3l. The thermal degradation of SA-MCC initiated at approximately 300 °C and continued with increasing temperature. The TGA of Zr-MOF particles exhibited an approximate 5% weight loss between 75

and 100 °C, which corresponds to the departure of guest molecules (chloroform or H₂O). Subsequently, a decrease occurred at around 200 °C, attributed to the dehydration of Zr₆O₄(OH)₄ nodes into Zr₆O₆. The final decrease near 350 °C to 400 °C is ascribed to the decomposition of organic linkers. Incorporating MOF nanoparticles into the SA matrix results in MOF-EMCC microcapsules with slightly higher decomposition temperatures, depending on the loading level. Notably, the onset of major weight loss shifts beyond 350 °C, indicating enhanced thermal stability. Additionally, a decrease in total weight loss is observed as MOF content increases, highlighting the thermal reinforcing effect of the nanoparticles.

Mechanical Properties of SA-MCC and MOF-EMCC Microcapsules. The mechanical properties of SA-MCC and MOF-EMCC microcapsules were evaluated, with the results presented in Figure 4e. The dried microcapsules had diameters of approximately 1 mm and masses ranging from 0.022 to 0.03 g. The bearing pressure of the SA-MCC sample is only 0.02 N, which is equivalent to 100 times its mass. In contrast, the loading pressure of the MOF-EMCC-60 sample reaches 5.2 N, which is 2×10^4 times its mass. The -NH₂ group in UiO-66-NH₂ not only strengthens CO₂ affinity but also stabilizes the MOF-alginate interface, contributing to the microcapsule's

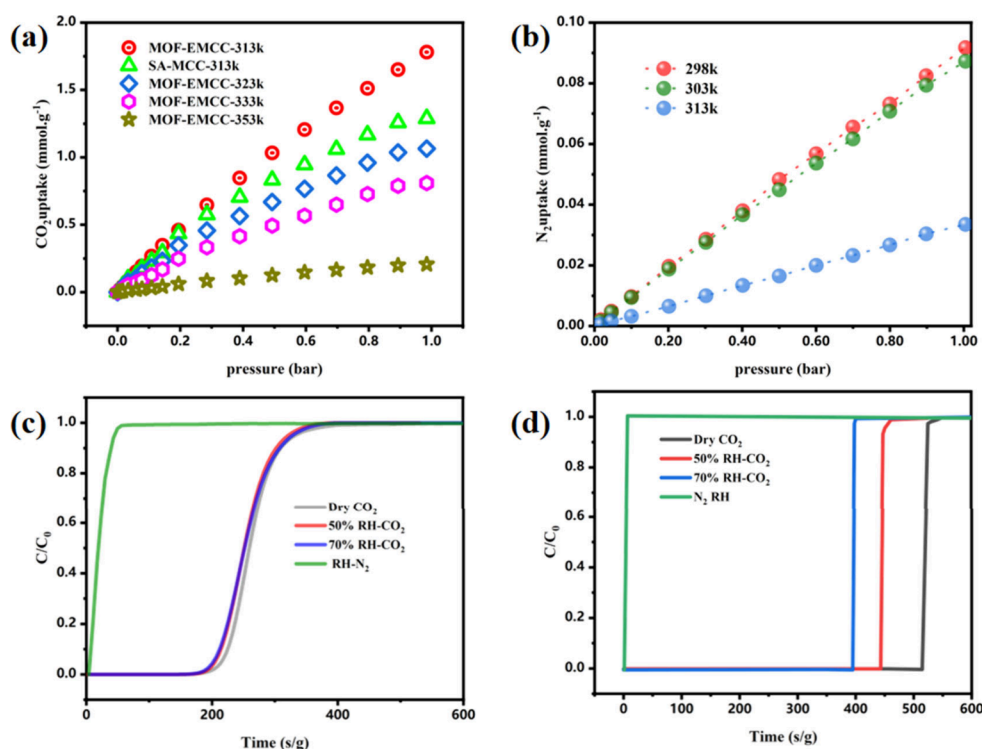


Figure 5. (a) CO_2 sorption isotherms at different temperatures, (b) N_2 sorption isotherms at different temperatures, (c) breakthrough curves of SA-MCC for CO_2/N_2 (15:85) mixture with different humidity, (d) breakthrough curves of MOF-EMCC for CO_2/N_2 (15:85) mixture with different humidity.

exceptional mechanical strength and humidity resistance (Figures 4e and 5d). These results far exceed the expected performance for carbon capture mediums, and indicate the strong potential of MOF-EMCCs for efficient CO_2 adsorption under practical conditions.^{32,33} The test results demonstrate a clear trend: as the MOF content increases, so does the mechanical strength of the microcapsules. This trend highlights the role of MOF nanoparticles as effective reinforcing agents that enhance the mechanical properties of the alginate matrix.

However, an increase in MOF content from 60 wt % to 70 wt % resulted in a decrease in the carrying capacity of the microcapsules. This reduction can be attributed to two factors. First, the issue of filler accumulation exceeding a certain threshold of MOF content leads to agglomeration due to self-adsorption or electrostatic interactions, which disrupts the uniform distribution within the matrix and diminishes the optimal effect. Second, the issue arises from interfacial problems between the MOFs material and the matrix. An excessive content of MOFs material can lead to excessively large interfacial gaps, weak adhesion with the matrix, and a lack of interfacial connectivity among the MOF particles themselves. These disruptions hinder the structural coherence of the composite, ultimately reducing the overall mechanical performance.

Gas Transport Properties. To gain a comprehensive understanding of the CO_2 mass transfer process in MOF-EMCC and to elucidate the efficient transport pathways facilitated by MOF (Figure 4a), we performed a series of density functional theory (DFT) simulations using VASP. These calculations enabled detailed insights into the diffusion channels illustrated in Figure 4b,c.

Figure 4a illustrates the CO_2 transport mechanism within the MOF-EMCC structure. The porous MOF channels

(highlighted in gold) embedded within the sodium alginate matrix act as low-resistance diffusion pathways for CO_2 molecules, while simultaneously restricting N_2 permeation. Panels b and c of Figure 4 display the DFT-calculated CO_2 diffusion pathways within the MOF-EMCC shell, and confirm reduced steric hindrance and more favorable adsorption configurations as a result of $-\text{NH}_2$ functionalization.

In Figure 4d, the diffusion energy barrier for CO_2 in MOF-EMCC (0.31 eV) is significantly lower than that in SA-MCC (0.81 eV), validating the enhanced mass transfer efficiency. This reduction aligns with the experimentally observed 38% increase in CO_2 adsorption capacity (Figure 5a). When MOF were incorporated into the sodium alginate matrix, the CO_2 adsorption energy for the MOF-EMCC shell was calculated at -0.61 eV, which is lower compared to -1.58 eV for the SA-MCC shell. This indicates a weaker CO_2 binding affinity in the MOF-EMCC, facilitating enhanced mass transfer of CO_2 . The results demonstrated that the diffusion energy barrier for CO_2 in MOF-EMCC was significantly lower than that in SA-MCC.

Notably, although the diffusion energy barrier for CO_2 in SA-MCC decreased under elevated pressure (1 bar), the corresponding reduction for MOF-EMCC was even more pronounced. This further reinforces the superior performance of MOF-EMCC, confirming that the incorporation of MOFs into the microcapsule shell substantially reduces CO_2 mass transfer resistance and boosts the carbon capture efficiency of the system.

Absorption and Recycling of CO_2 in Microcapsules.

The CO_2 absorption capacity of the microcapsules was determined by measuring the corresponding adsorption isotherms at various temperatures. Performing CO_2 adsorption using a CO_2/N_2 swing at 313 K, which represents the typical temperature for carbon dioxide capture in power plants. As

shown in Figure 5a, the microcapsules demonstrated excellent CO₂ capture capacity at 313 K. The SA-MCC exhibited a CO₂ capture capacity of 1.29 mmol/g, while the MOF-EMCC showed a significantly higher capacity of 1.78 mmol/g, representing an approximate 38% enhancement. The –NH₂ group acts as a Lewis basic site, enabling dipole–quadrupole interactions and hydrogen bonding with CO₂ (enhancing selectivity and capacity), while its porous framework reduces diffusion resistance.

The CO₂ adsorption performance of [Bmim][Ac] was tested under the same experimental conditions (see Figure S7). The results demonstrate that the ionic liquid alone achieved a CO₂ capacity of 0.99 mmol/g (313 K, 1 bar), while the MOF-EMCC microcapsule system exhibited an 80% improvement (1.78 mmol/g). This highlights the critical role of the MOF-doped shell in enhancing mass transfer and overall capture efficiency. The acetate anion ([Ac][−]) facilitates chemical absorption via carboxylate–CO₂ reactions, providing high intrinsic capacity but limited kinetics due to viscosity. Encapsulation mitigates [Bmim][Ac]'s kinetic limitations by increasing interfacial area, while UiO-66-NH₂ channels accelerate CO₂ transport to the ionic liquid core (validated by DFT simulations in Figure 4d). Additionally, the CO₂ capture performance of MOF-EMCC was assessed at various temperatures. At 323 K, 333 K, and 353 K, the MOF-EMCC microcapsules exhibited CO₂ capture capacities of 1.06 mmol/g, 0.81 mmol/g, and 0.21 mmol/g, respectively. These results highlight a strong temperature dependence, with superior performance at lower temperatures and a notable decline in adsorption at elevated temperatures.

Through the conduct of CO₂ adsorption and desorption cycles at temperatures of 313 K and 353 K, the net CO₂ capture capacity was calculated to be 1.57 mmol/g. Meanwhile, the isosteric heats of adsorption (Q_{st} = 13.2 kJ/mol for MOF-EMCC vs 15.8 kJ/mol for UiO-66-NH₂) confirm that the MOF-doped shell optimized the balance between adsorption capacity and regeneration efficiency (Figures S8 and S9). This reduction aligns with the lower CO₂ binding energy observed in DFT simulations (Figure 4d), ensuring energy-efficient desorption at mild temperatures (313 K–353 K). Collectively, these results confirm that MOF-EMCC exhibits important characteristics of low CO₂ capture energy consumption. A comparative analysis summarized in Table 1 shows that MOF-

polymers. As a result, microcapsules doped with MOFs exhibit superior mass transfer capabilities and enhanced CO₂ capture effects while maintaining high CO₂ selectivity.

Experimental results confirmed that MOF-EMCC displayed exceptional CO₂ capture performance. The adsorption isotherms for both CO₂ and N₂ are presented in Figure 5a,b. Notably, at 298 K, 303 K, and 313 K, the N₂ adsorption capacities of MOF-EMCC are 0.09 mmol/g, 0.08 mmol/g, and 0.03 mmol/g, respectively. At 313 K, the N₂ capacity is just 1.6% of the CO₂ adsorption capacity (1.78 mmol/g), which highlights the remarkable selectivity of the system.

As previously discussed, MOF-EMCC exhibits exceptional CO₂ capture capacity, suggesting its potential application in capturing CO₂ from flue gases. To verify the feasibility of this application, dynamic fixed-bed breakthrough experiments were conducted on a CO₂/N₂ (15:85) gas mixture, simulating coal flue gas, at 313 K and 1 atm pressure. The breakthrough curve for MOF-EMCC, shown in Figure 5d, reveals that N₂ initially reaches the breakthrough point at 3.1 s/g (breakthrough point, where the concentration of the effluent gas reaches 5% of the influent concentration on the breakthrough curve), and 11.6 s/g at the saturation point (where the concentration of the effluent gas reaches 95% of the influent concentration on the breakthrough curve). The curve also indicates that CO₂ reaches the breakthrough point at 516.4 s/g and the saturation point at 551 s/g, with a saturation duration of 34.6 s/g and a breakthrough time of 539.4 s/g.

Comparatively, under identical conditions, the CO₂ capture performance of SA-MCC in coal flue gas was also assessed. As shown in Figure 5c, N₂ and CO₂ reach their respective breakthrough points at 13.8 s/g and 185 s/g, and saturation points at 38.6 s/g and 361 s/g. Further, the saturation duration for CO₂ is 176 s/g and the breakthrough time is 347.2 s/g. These experiments reiterated that MOF-EMCC provides a substantially higher CO₂ capture capacity than SA-MCC, with a breakthrough time of 539.4 s/g compared to 347.2 s/g for SA-MCC, and a significantly faster CO₂ capture rate (approximately five times that of SA-MCC = saturation duration of SA-MCC/saturation duration of MOF-EMCC). Moreover, despite the presence of water vapor, MOF-EMCC continues to demonstrate excellent CO₂ capture capacity and rate. While DFT simulations suggest weaker CO₂ binding in the shell may contribute to this enhancement, the experimental data conclusively demonstrate the efficacy of the hierarchical pore structure in reducing mass transfer resistance.

From the breakthrough data, the dynamic CO₂/N₂ selectivity and CO₂ adsorption capacity were quantitatively evaluated. The separation factor for SA-MCC was calculated as 12, whereas MOF-EMCC exhibited a significantly higher value of 28, demonstrating a 133% improvement in selectivity. Furthermore, the dynamic CO₂ uptake of SA-MCC reached 0.87 mmol/g, which is 12% lower than the pure ionic liquid ([Bmim][Ac]) due to shell-induced mass transfer limitations. In contrast, MOF-EMCC achieved a dynamic capacity of 1.62 mmol/g under identical conditions, surpassing both SA-MCC and the parent ionic liquid by 86% and 45%, respectively.

Lastly, the cyclic stability of MOF-EMCC was evaluated over 20 adsorption–desorption cycles, demonstrating consistent performance with negligible degradation in both structure and CO₂ capacity (Figure S6). These findings confirm that MOF-EMCC microcapsules not only offer superior selectivity and capacity but also provide robust

Table 1. Performance Comparison of MOF-EMCC with Ionic Liquids (CO₂ Capacities (g of CO₂ per g of Total Microcapsule Mass))

Material	CO ₂ Capacity(g/g)	CO ₂ /N ₂ Selectivity	Condition (bar/K)	Ref
[bmim][DCA]	0.04	15	14.2/313	34
[bmim][NO ₃]	0.052	20	29.1/313	35
[BMP][Tf ₂ N]	0.056	22	19.3/313	36
[C ₈ mim][PF ₆]	0.07	25	29.5/313	35
MOF-EMCC (this work)	0.078 (1.78 mmol/g)	28	1/313	

EMCC achieves a CO₂ capacity of 1.78 mmol/g (313 K, 1 bar) and a CO₂/N₂ selectivity of 28, outperforming many reported ionic liquid systems in both uptake capacity and kinetics.

The physical and chemical properties of the microcapsule shell material are pivotal in determining its CO₂ capture capacity. The crystalline porous architecture of MOFs facilitates more efficient molecular diffusion than conventional

operational stability, making them highly suitable for practical CO₂ capture from flue gas.

CONCLUSION

Carbon capture microcapsules efficiently integrate the high specific surface area of solids with the high CO₂ selectivity of carbon capture solvents to facilitate rapid and energy-efficient absorption and release of CO₂. Future developments in this field may benefit from the incorporation of well-established carbon capture solvents and insights from gas separation membranes technologies to guide the selection of shell materials and adsorbents. However, early encapsulation strategies, such as the seminal silicon-shell system for carbonate solutions reported by Vericella et al.,²⁸ demonstrated that the permeability of the microcapsule shell material is often limited, presenting significant resistance to CO₂ mass transfer and thereby restricting kinetics and the broader advancement of the technology. Our approach fundamentally addresses this key limitation. The novel carbon capture microcapsules introduced in this study innovatively show that incorporating engineered MOF channels (UiO-66-NH₂) within a flexible, cross-linked biopolymer (sodium alginate) shell overcomes the mass transfer barrier inherent in dense shell materials like silicon. This synergy not only significantly enhances CO₂ mass transfer kinetics (~5x faster capture rate) and capture capacities (1.62 mmol/g vs pure IL's 0.99 mmol/g under flue gas simulant), but also concurrently improves mechanical properties (withstanding loads up to 20,000x their mass), addressing another critical weakness of earlier rigid-shell designs. The fabrication of MOF-EMCC using high-frequency EVT further provides a scalable and cost-effective route compared to intricate microfluidic techniques often used previously, overcoming production challenges.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c10814>.

Experimental setup schematics; equipment photographs; microcapsule formation mechanisms; optical/SEM characterization images; cyclic stability data; adsorption isotherms; breakthrough curves; heat analysis (PDF)

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Notes

The authors declare no competing financial interest.

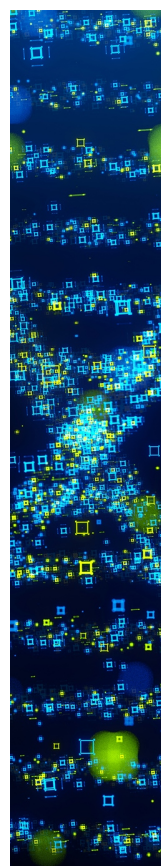
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